

Student number:



EEE3094S 2024: Class Test 1 - Memo

- The test is open book
- The supporting information for all questions and formula sheet are attached at the end of this paper. You may remove them for easy access.
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write “see attached pages” by the question) and make sure they are marked with your student number.
- You may do your diagrams in pencil but please ensure that your lines are clear.
- There are 3 questions worth a total of 56 marks. The maximum mark you can score for the test is 50 though.
- You have 90 minutes to complete the test.

Question 1

The aim of this question is to create a block diagram describing altitude control in the enhanced drone.

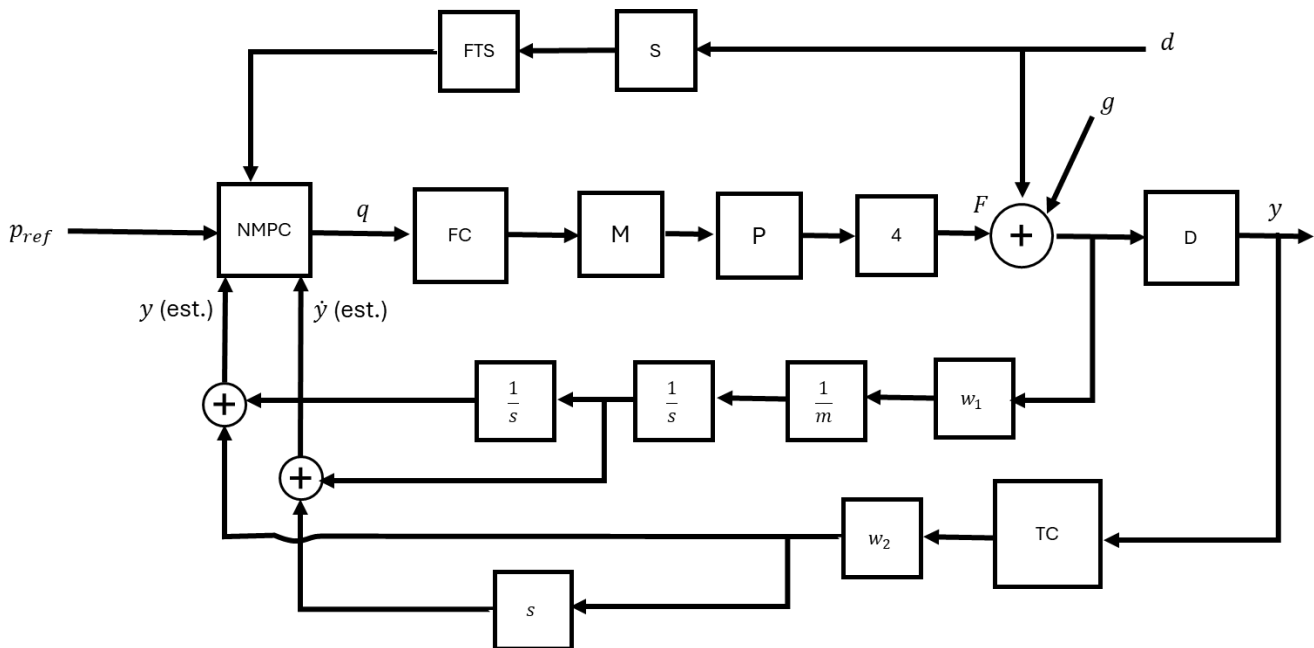
Draw a block diagram describing height control of the drone. Your diagram should include the following subsystems: (A large space is provided on the next page) [20 marks]

- The model-predictive controller, NMPC
- The flight controller, FC
- The motors (M)
- The propellers (P)
- The drone body, B
- The plastic shell, S
- The accelerometer (modelled as described)
- The tracking camera, TC
- The force/torque sensor, FTS

Indicate the following signals on your diagram:

- Goal position p_{ref}
- Position input to flight controller q
- Thrust input to drone F
- Gravity g
- Collision disturbance d
- Altitude y

Solution



Question 2 [18]

This question develops an open-loop model of the basic drone.

- a) The relationship between the input EMF $e(s)$ and thrust $F(t)$ produced by the motor-propeller subsystem can be approximated by a first-order model. Give this model as a transfer function and explain why it is an acceptable approximation of this system's behaviour. (Answer in terms of the given variables, rather than numerical values for the parameters.) [5 marks]
- b) Following on from a, give an open-loop transfer function model for the drone relating $e(t)$ to the vertical **velocity** $\dot{y}(t)$, which incorporates the first-order approximation of the motor and propeller. (Answer in terms of the given variables, rather than numerical values for the parameters.) [2 marks]
- c) Your model in b should be a second-order system, but it is possible to simplify it even further, down to a first-order transfer function. Based on the given vertical velocity response of the drone to a unit step in the input $e(s)$, explain why this is an acceptable simplification. (Hint: are there any specifically second-order features in this step response?)

Suggest a suitable first-order model of the form $\frac{A}{Ts+1}$ **based on the step response**. You may annotate the copy of the plot to show important features that support your reasoning [4 marks]

- d) Consider the given Bode plot of the system. Is there a range of frequencies where you would say a first-order assumption holds? (Answer to the nearest order of magnitude/ multiple of 10). Explain. If so, use features of the response within this range to confirm the first-order model you obtained in part c. [4 marks]
- e) Does the first-order model you obtained from observing the system's step response and frequency response seem correct considering the measured parameters of the drone's motors, propeller and body? [3 marks]

Solutions:

- a) The motor-propeller subsystem model consists of the motor's mechanical and electrical components, and the propeller.

The differential equation relating the motor's shaft velocity to the input torque gives the transfer function

$$\frac{\dot{\theta}(s)}{\tau(s)} = \frac{1}{Js + b} = \frac{\frac{1}{b}}{\frac{J}{b}s + 1}$$

When cascaded with the propeller (modelled as the scalar k_p) and the described transfer function relating the torque to the EMF, this gives the model

$$\frac{F(s)}{e(s)} = \frac{k_T k_p \frac{1}{bR}}{\left(\frac{J}{b}s + 1\right) \left(\frac{L}{R}s + 1\right)}$$

It is stated that the electrical time constant $\frac{L}{R}$ is much smaller than the mechanical constant $\frac{J}{b}$, meaning that the pole created by the electrical elements will be dominated by the pole representing the mechanical elements. We can therefore reduce the order of this model by replacing the pole term $\left(\frac{L}{R}s + 1\right)$ with its gain, giving the first-order model

$$\frac{F(s)}{e(s)} = \frac{k_T k_p \frac{1}{bR}}{\frac{J}{b}s + 1}$$

- b) The complete open-loop model of the drone is obtained by multiplying the model of the motor and propeller with that of the drone body.

From the given differential equation, the effect of the thrust on the drone's vertical velocity can be modelled by the transfer function

$$\frac{\dot{y}(s)}{F(s)} = \frac{1}{ms + a} = \frac{\frac{1}{m}}{\frac{m}{a}s + 1}$$

When combined with the motor-propeller model, the open-loop transfer function of the drone is

$$\frac{\dot{y}(s)}{e(s)} = \frac{4k_T k_p \frac{1}{bLm}}{\left(\frac{J}{b}s + 1\right)\left(\frac{m}{a}s + 1\right)}$$

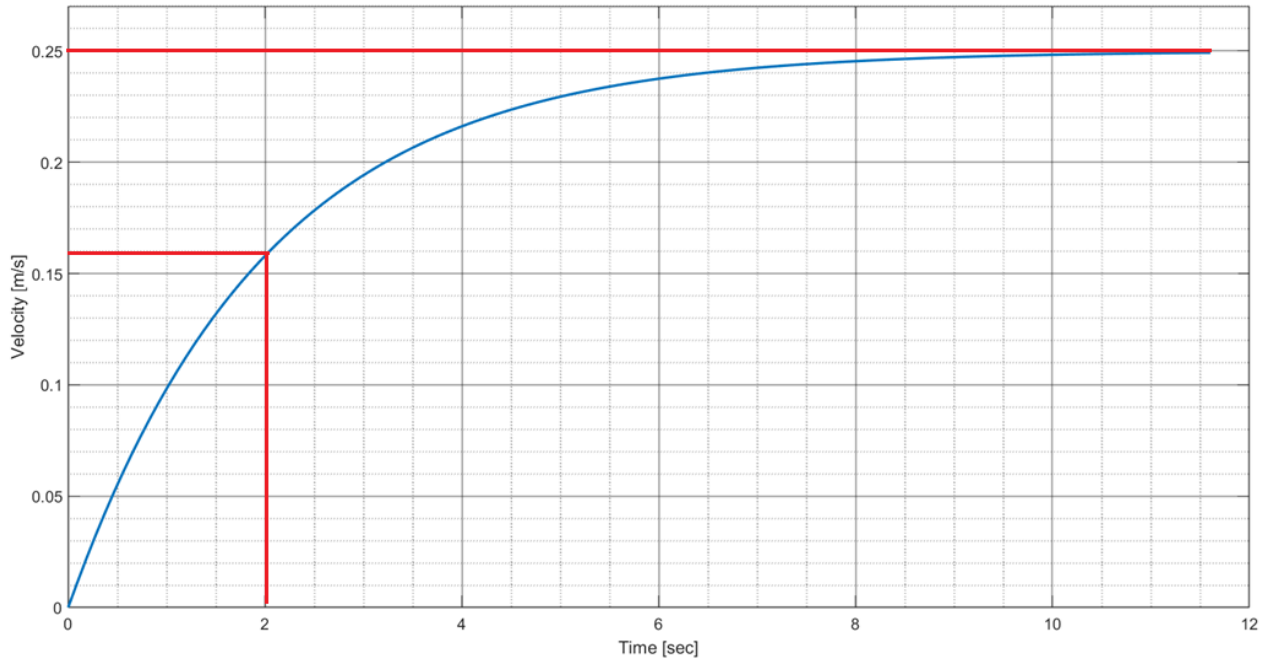
(The transfer function for the motor-propeller subsystem is scaled by four, because there are four of them.)

- c) The only characteristically second-order behaviour would be oscillation, as this requires a pair of complex poles. There is no visible oscillation in this step response, so it can potentially be represented by a first-order model.

The gain A is given by the final value, which is around 0.25.

The time constant will be the time where the plot reaches about 0.63A, or 0.16. This occurs at 2 seconds, so $T = 2$.

The model is therefore $G(s) = \frac{0.25}{2s+1}$



- d) In a first-order response, we would expect to see a corner frequency with no resonant peak, followed by a 20 dB/dec. roll-off. We see this for frequencies below about 100 rad/s.

The low-frequency gain on the Bode plot is about -10 dB, or 0.3. Considering the scale of the Bode plot compared to the step response, this seems reasonably close to the gain of 0.25 obtained before.

The corner frequency of the slowest pole can be found by finding where the low and high frequency asymptotes (in this case, the -20 dB/dec. asymptote) intersect, or the point where the phase is -45 degrees. This happens at a frequency of 0.5 rad/s, which agrees with the time constant of $\frac{1}{\omega_c} = 2$ obtained before.

- e) The gain of the model is given by the parameters $4k_T k_p \frac{1}{bLm}$. Subbing in these values from the information sheet gives $\frac{4(0.008)(0.0025)}{(0.0062)(0.13)(0.4)} = 0.25$, confirming the gain obtained from the step response.

The poles in the original model had time constants equal to the mechanical constant, and the ratio $\frac{m}{a}$, so the slower of these should give the observed time constant. The mechanical time constant is given to be 0.0013, while $\frac{m}{a} = \frac{0.4}{0.2} = 2$, which matches the observed value.

The proposed first-order model therefore agrees with the known parameters of the drone.

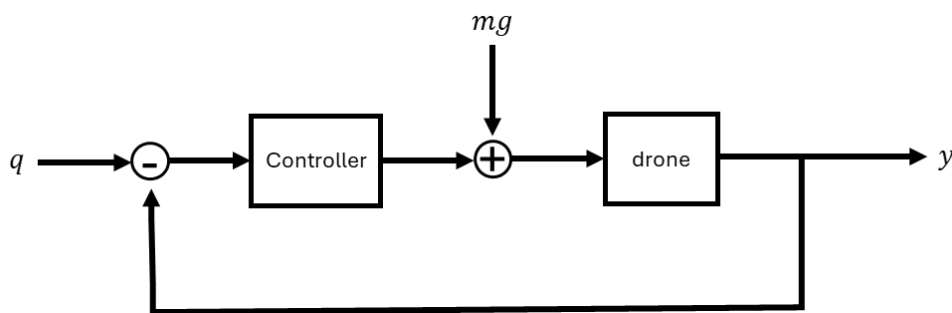
Question 3 [18]

This question uses the drone model derived in the previous question to design a simple flight controller to regulate altitude. (Note: the altitude model is just the integral of the vertical velocity model you derived before.) If you had trouble getting a model, you can use the transfer functions below to represent the relationship between altitude and EMF:

$$\text{Full-order model: } G(s) = \frac{0.25}{s(0.00014s+1)(0.0013s+1)(2s+1)}$$

$$\text{Reduced-order approximation: } G_r(s) = \frac{0.25}{s(2s+1)}$$

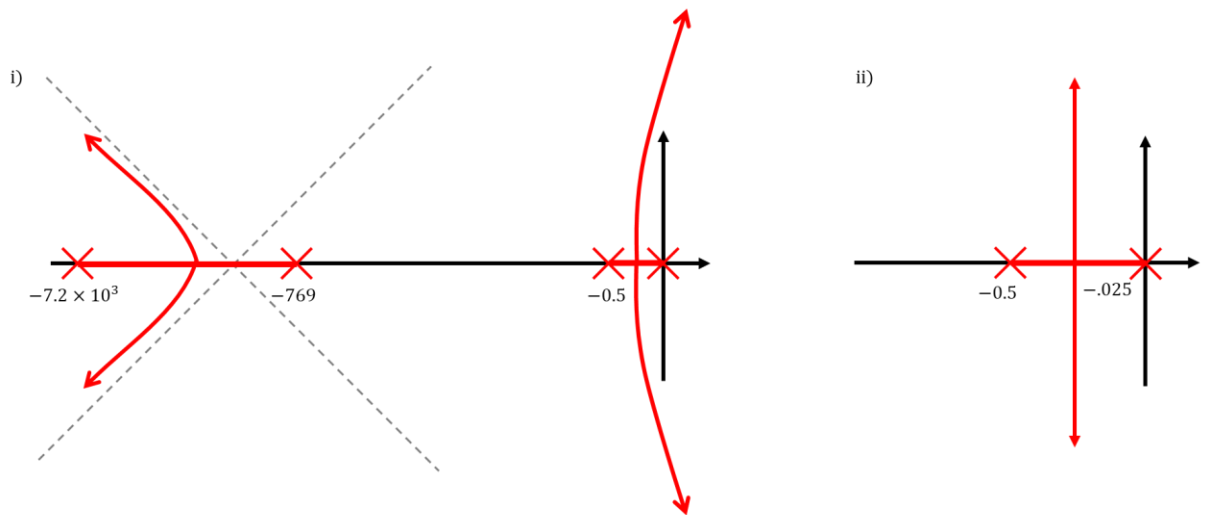
You want to design a simple flight controller that will receive the desired altitude input $q(s)$ from the NMPC and adjust the altitude of the drone to match it. You can assume that the drone is flying in open skies (no collisions) and that the sensing of the altitude is perfect, giving the simplified closed-loop system shown in the block diagram below.



- Sketch rough root locus plots for the drone, modelling the relationship between height y and EMF e using (i) the full fourth-order model, (ii) the reduced-order model. Hint: this is a *rough* plot, so your drawing does not need to be to scale. [6 marks]
- Comparing the root locus plot in (i) to the one in (ii), is there any unique information that you get from the full-order model that is lost when you move to the simplified one? (Hint: consider the stability and transient response characteristics of the closed-loop system) [2 marks]
- BONUS: Explain why the closed-loop system will track the altitude perfectly, without disturbance from gravity, regardless of gain. What would be required to achieve perfect tracking of the vertical velocity? [4 marks]
- If you want to use a proportional controller, how large would the gain need to be for the altitude to track a *velocity* setpoint with less than 10% error? [2 marks]
- Based on the given Nyquist plot of the open-loop altitude model's frequency response, would you say that it is possible to achieve velocity tracking with less than 10% error without making the system unstable? Explain. [4 marks]

Solutions:

- This model has four poles and no zeros so there must be four asymptotes. The poles are at 0, -0.5, -770, and -7.2×10^3 . This gives a centroid at about -2000. The angles of the asymptotes will be $\pm \frac{\pi}{4}$, and $\pm \frac{3\pi}{4}$.
 - This model has two poles and no zeros, so two asymptotes. The poles are at 0 and -0.5, so the centroid will be at -0.25. The angles of the asymptotes are $\pm \frac{\pi}{2}$.



- b) The full-order model shows that the poles cross into the right-half plane, therefore there are some gains that will cause the closed-loop system to become unstable. This is not evident in the root locus of the reduced-order model.
- c) The position model is a type 1 system (has one integrator in the forward path) so the tracking error for position inputs will be reduced to zero.

Likewise, zero-order input disturbances such as gravity will be rejected completely.

To track velocity inputs perfectly, the controller would need to raise the type number of the system to 2, i.e. introduce another integrator to the forward path.

- d) The magnitude of $G(s)$ is 0.25, so the gain of the closed-loop system is $\frac{0.25K}{1+0.25K}$. (This is true regardless of whether the full- or reduced-order model is considered).

$$\frac{0.25K}{1+0.25K} \geq 0.9 \therefore 0.025K \geq 0.9 \therefore K \geq 36$$

- e) For the system to meet this tracking specification without becoming unstable, it would need to have a gain margin of at least 36, since that is how much the proportional controller will increase the gain by.

The open-loop system has no poles in the right-half plane, so the Nyquist stability criterion indicates that any clockwise encirclements of -1 will lead to unstable closed-loop poles ($N = Z - 0$).

The detailed Nyquist plot shows that the value where it intersects the negative real axis must be extremely close to zero, however, so the gain can be increased well beyond the suggested value without causing an encirclement.

The system should therefore be able to satisfy this tracking specification.

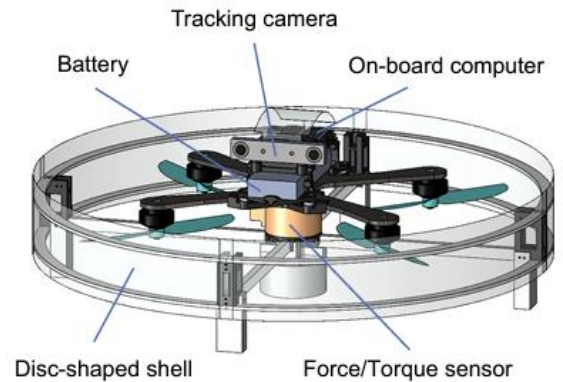
System Information

All questions in this test are based on information in the following document. The content has been adapted and simplified to fit the context of a class test, but the ideas are consistent with real research published this year in the scientific journal 'Nature Communications'.

Background

For aerial robots (drones) to be useful in a variety of environments, they need to be able to navigate obstacles safely. Generally, this involves avoiding these obstacles entirely, but in the case of some compliant (bendy) objects such as leaves or soft branches and twigs, a better option might be to just push past them the way animals typically do.

One of the reasons this is easy for animals is because of their ability to sense collisions with objects anywhere on their bodies. Researchers are trying to give drones the same ability by surrounding them in bouncy plastic shells equipped with force/torque sensors (Aucone, et al. 2024)¹.



Altitude control of the drone (Question 1)

Control of the drone's position is accomplished in two stages: at the higher level, a nonlinear model-predictive controller (NMPC) uses a simulation of the drone's dynamics to determine the desired position of the drone, q . The lower-level flight controller (FC) converts this command to voltages applied to each of the drone's four brushless DC motors (M), causing it to rotate at the required velocity.

Each motor drives a propeller (P) causing thrust to be generated. The combination of these thrusts gives the total actuator force F acting on the drone's body (B), causing it to move to an altitude y . The body is also acted upon by gravity (g), and disturbances (d) caused by collisions with obstacles.

During vertical movement, it can be assumed that the drone body is parallel to the ground, with all motors receiving the same commands and turning at the same velocity.

The altitude and upward velocity ($\dot{y}$) of the drone are detected by two sensors:

1. An accelerometer, which can effectively be thought of as a reading of the *force acting on the drone's body*, scaled by $\frac{1}{m}$, where m is the drone's mass, and integrated.
2. A tracking camera (TC). The tracking camera gives a position approximation based on landmarks, which is then differentiated to estimate the velocity.

The force created by collision disturbances is sensed by a force/torque sensor (FTS), which works by measuring the deflection of the plastic shell (S) produced by d .

The NMPC takes multiple inputs:

- 1) The ultimate goal position of the drone, p_{ref} .

¹ Aucone, Emanuele, et al. "Synergistic morphology and feedback control for traversal of unknown compliant obstacles with aerial robots." *Nature Communications* 15.1 (2024): 2646.

- 2) The approximate position of the drone, which can be simplified as a weighted combination of the accelerometer and camera's position estimates. These are given weights w_1 and w_2 , respectively.
- 3) The approximate velocity of the drone, which can be simplified as a weighted combination of the accelerometer and camera's velocity estimates. These are given weights w_1 and w_2 , respectively.
- 4) The reading from the force/torque sensor.

Open-Loop model of the drone (Question 2)

The open-loop dynamics of the drone are a combination of the dynamics of the four motors and propellers, and the drone body. We will only consider vertical motion.

The input to the drone is the EMF $e(t)$ applied to the motors by the flight controller.

Motor Model

The relationship between torque input $\tau(t)$ and shaft velocity $\dot{\theta}(t)$ for each motor is given by the differential equation

$$J\ddot{\theta}(t) = -b\dot{\theta}(t) + \tau(t)$$

where J is the inertia of the motor, and b is a damping coefficient capturing the effect of friction.

The torque developed by the motor is related to the input current $i(t)$ by a scalar multiplier, the constant k_T .

The current is related to the input EMF by the transfer function

$$\frac{i(s)}{e(s)} = \frac{1}{R + sL} = \frac{\frac{1}{R}}{1 + s\frac{L}{R}}$$

where R and L are the resistance and inductance of the winding.

Note: it is important for friction in the motor to be very low for efficient operation. For this reason, the ratio $\frac{J}{b}$ will generally be very large compared to the electrical constant $\frac{L}{R}$ in a motor that is functioning properly.

Propeller Model

The conversion from motor shaft velocity to aerodynamic thrust performed by the propeller can be represented by a scalar, $k_p = \frac{F(t)}{\dot{\theta}(t)}$.

Drone Body Model

The upward velocity of the drone is related to the upward force applied by the differential equation

$$m\ddot{y}(t) = F(t) + a\dot{y}(t)$$

where m is the mass of the drone and a is a constant capturing the effect of aerodynamic drag.

Drone Parameters

Weighing the drone gives mass $m = 0.4$ kg

The datasheet of the drone's motors gives the following parameters:

Torque constant $k_T = 0.008$ Nm/A

Shaft inertia $J = 8 \times 10^{-6}$ kg.m²

Mechanical time constant $\frac{J}{b} = 1.3$ msec

Inductance $L = 1.8 \times 10^{-5} \text{ H}$

Resistance $R = 0.13 \Omega$

From experiments, the propeller constant is estimated to be around $k_p = 0.0025 \text{ Nm per rad/s}$

The aerodynamic drag constant is estimated to be around $a = 0.2 \text{ N per m/s}$.

Open-Loop Response of Drone

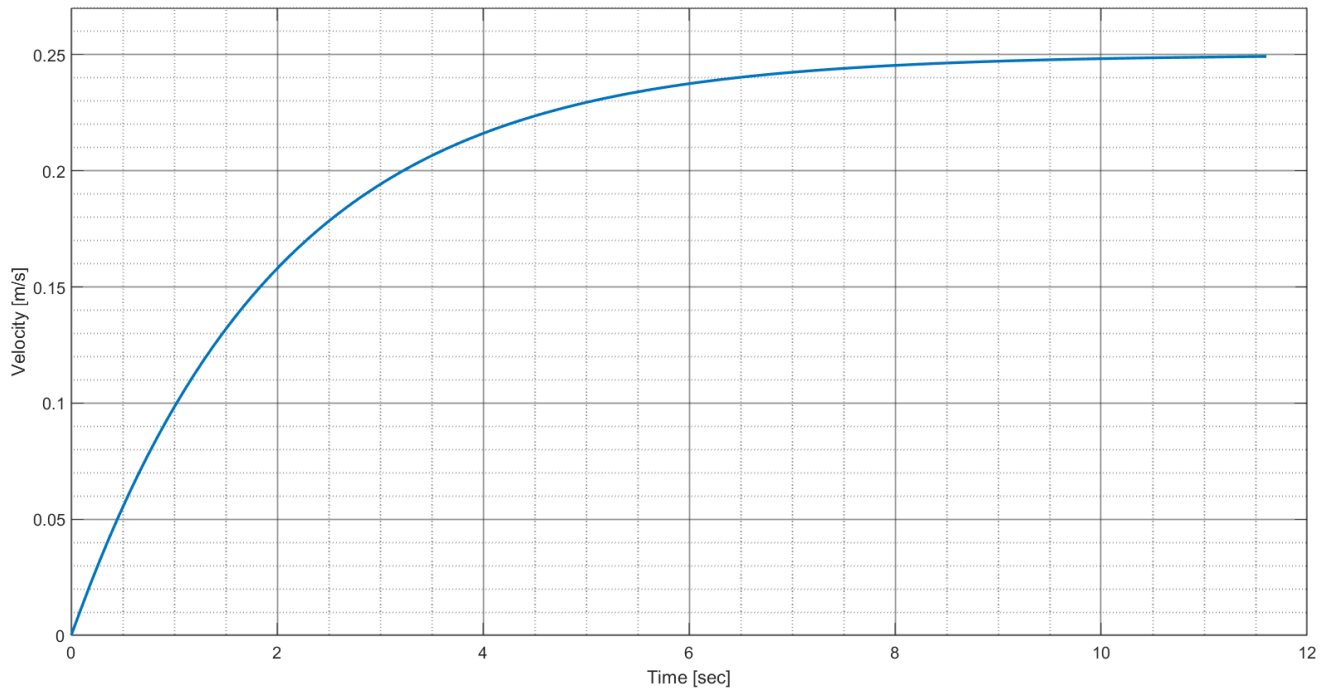


Figure 1: Vertical velocity of the drone in response to a unit step applied in the input EMF to the motors.

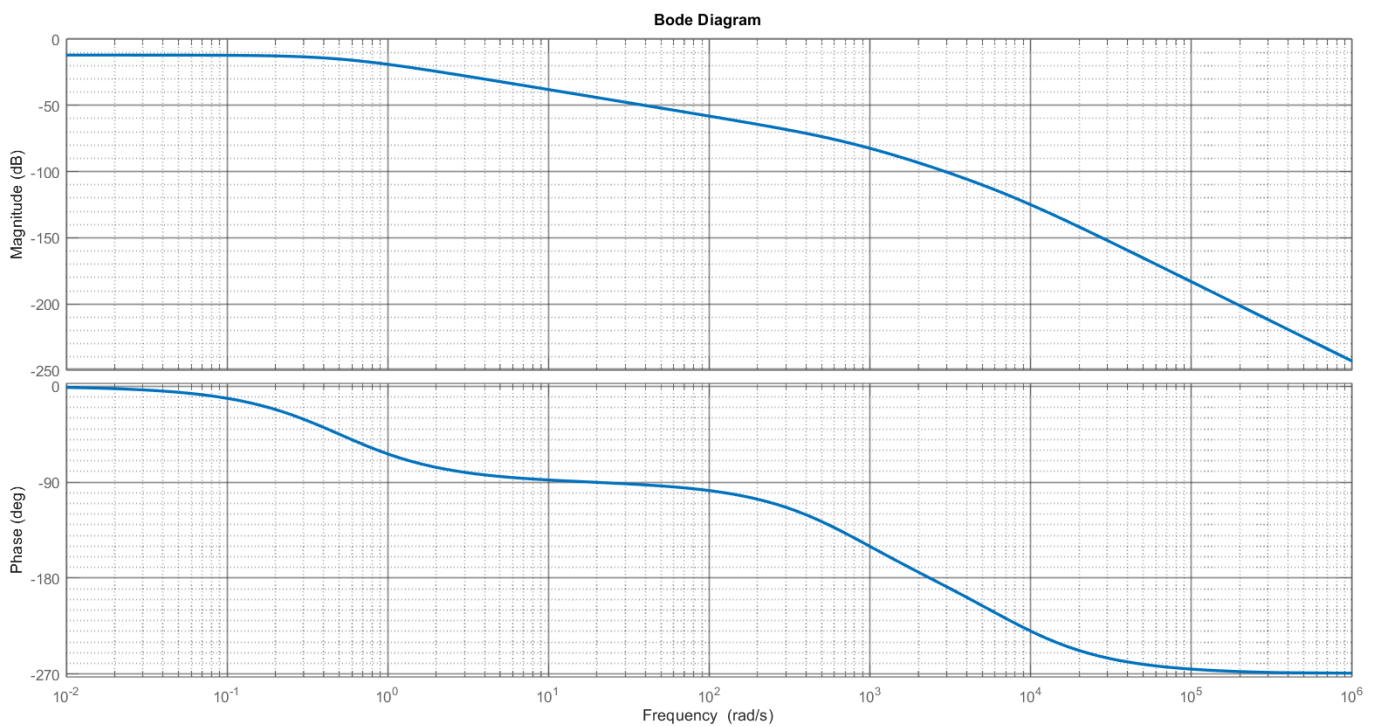


Figure 2: Open-loop frequency response of the drone's vertical velocity on a Bode plot

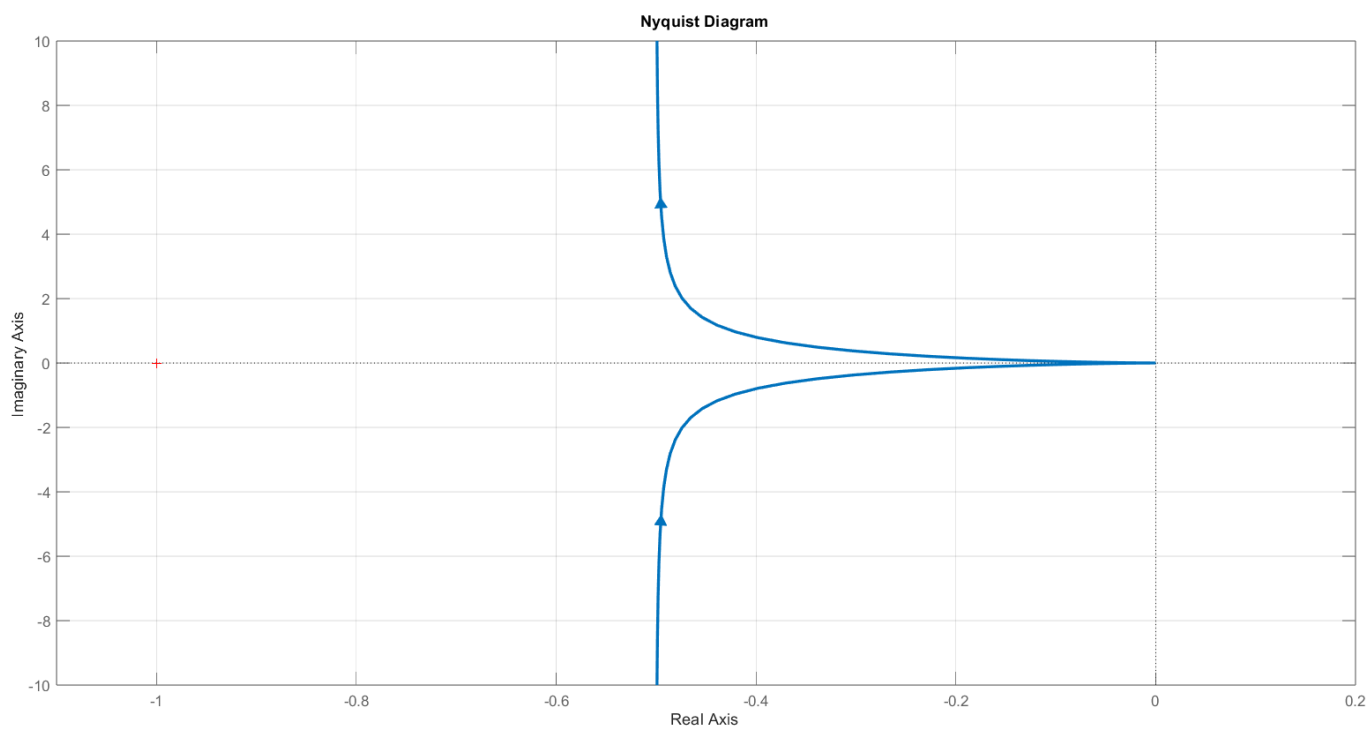


Figure 3: Open-loop frequency response of the drone's altitude on a Nyquist plot

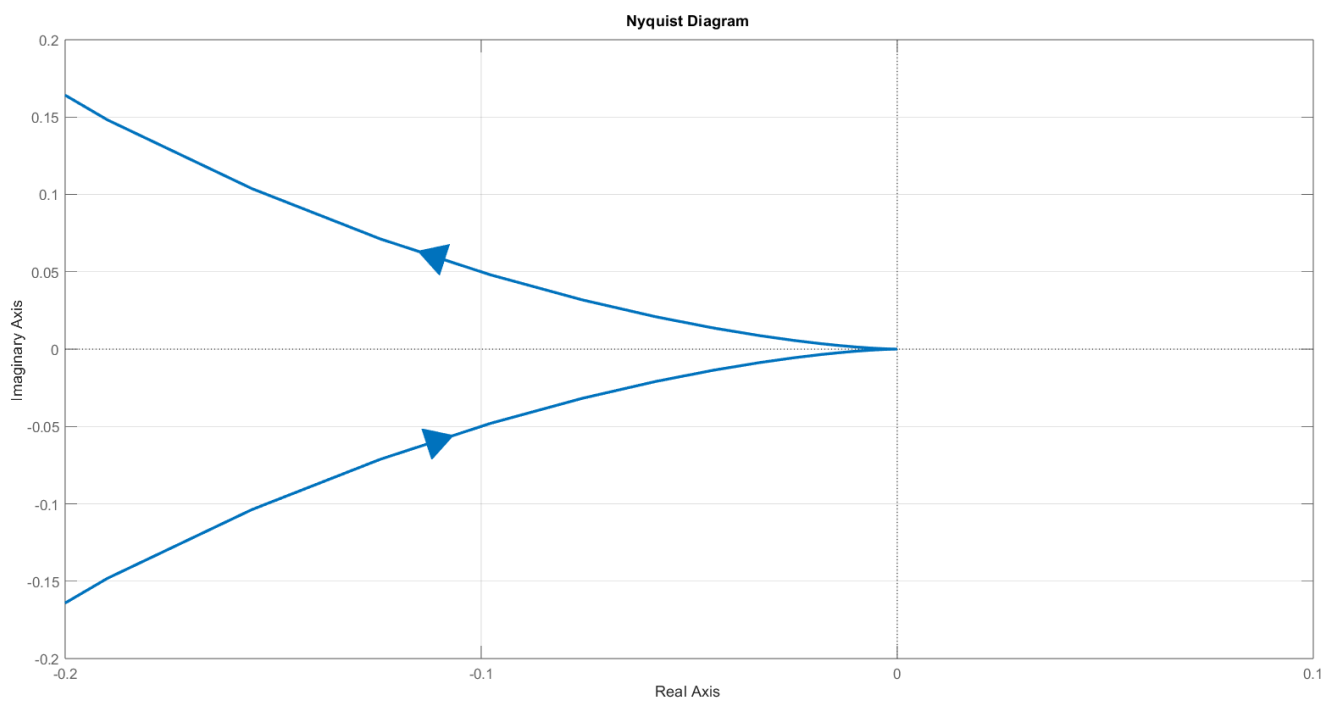


Figure 4: Detail view of the Nyquist plot in Figure 3.